FIGURE 2 CONTINUED · EXISTING MECHANISTIC EVIDENCE · SHEET 1/2

Mechanistic explanations and their corresponding measurements

Companion to manuscript Figures 2–3. Existing mechanistic evidence explains specific contributions and motivates measurements; the common gain/recovery panels do not causally partition all three cue-duration curves.

Model / candidate mechanism	Observed evidence	Corresponding measurements
Wave-pinning A wider incumbent and conserved-pool competition make challenger growth and post-cue survival depend on spatial reorganization, even when initial local activation gain changes little.	At L15, width-3 pulses of 20/60 native units collapse/retain the challenger. At L39, fourfold inactive diffusivity rescues the native 5120-unit fixed-support failure. Gain and local deficit repair remain near-null with size.	Measure incumbent extent and excess signal, early challenger growth, boundary retreat and both sites after withdrawal. Pair local activity with substrate-recovery or redistribution measurements; distinguish peak height from region expansion.
Goryachev A taller incumbent sequesters conserved material, reducing inactive concentration at the challenger while increasing the incumbent's measured response to specified damage. Initiation and incumbent restoration are coupled contributions.	For L19→39, fixed-support initial gain falls to 0.654 and common active-loss repair rises 3.439-fold. Separate L9→39 inactive-bolus removal/return gives a 15.71-fold increase in immediate reaction per deficit.	Measure incumbent peak, physical extent and integrated signal; measure challenger activation under characterized input and available substrate. Use a paired small activity-loss/sham assay to distinguish immediate local restoration from subsequent recovery.

Inference scope

Wave-pinning: Withdrawal examples and an evolved fixed-dose mobility intervention support transport/domain competition. They do not prove constant front speed or quantify a new duration threshold.

Goryachev: Native input multiplies inactive concentration; equal cue peaks therefore differ in biochemical activation. Repair is perturbation- and clock-specific. Neither contribution supplies an exhaustive or additive allocation of reversal duration.

Source: mechanism_crosswalk_v2.json · Exact assays, gates and source citations accompany the full legend.